

An LLM-Powered Cooperative Framework for Large-Scale Multi-Vehicle Navigation

Yuping Zhou

The Hong Kong University of Science and Technology
(Guangzhou)
Guangzhou, Guangdong China
yzhou169@connect.hkust-gz.edu.cn

Jindong Han

The Hong Kong University of Science and Technology
(Guangzhou)
Guangzhou, Guangdong China
jindong.han@sdu.edu.cn

Siqi Lai

The Hong Kong University of Science and Technology
(Guangzhou)
Guangzhou, Guangdong China
slai125@connect.hkust-gz.edu.cn

Hao Liu*

The Hong Kong University of Science and Technology
(Guangzhou)
Guangzhou, Guangdong China
The Hong Kong University of Science and Technology
Hong Kong SAR, Hong Kong
liuh@ust.hk

Abstract

The rise of Internet of Vehicles (IoV) technologies is transforming traffic management from isolated control to a collective, multi-vehicle process. At the heart of this shift is multi-vehicle dynamic navigation, which requires simultaneously routing large fleets under evolving traffic conditions. Existing path search algorithms and reinforcement learning methods struggle to scale to city-wide networks, often failing to capture the nonlinear, stochastic, and coupled dynamics of urban traffic. To address these challenges, we propose *CityNav*, a hierarchical, LLM-powered framework for large-scale multi-vehicle navigation. *CityNav* integrates a global traffic allocation agent, which coordinates strategic traffic flow distribution across regions, with local navigation agents that generate locally adaptive routes aligned with global directives. To enable effective cooperation, we introduce a *cooperative reasoning optimization* mechanism, in which agents are jointly trained with a dual-reward structure: individual rewards promote per-vehicle efficiency, while shared rewards encourage network-wide coordination and congestion reduction. Extensive experiments on four real-world road networks of varying scales (up to 1.6 million roads and 430,000 intersections) and traffic datasets demonstrate that *CityNav* consistently outperforms nine classical path search and RL-based baselines in city-scale travel efficiency and congestion mitigation. Our results highlight the potential of LLMs to enable scalable, adaptive, and cooperative city-wide traffic navigation, providing a foundation for intelligent, large-scale vehicle routing in complex urban environments. Our project is available at <https://github.com/usail-hkust/CityNav>.

* Corresponding author.



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CCS Concepts

• Computing methodologies → Multi-agent planning.

Keywords

internet of vehicles, vehicle navigation, large language model, multi-agent control

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1 Introduction

The rapid development of Internet of Vehicles (IoV) [18, 36] technologies is transforming traffic control from an isolated activity into a collective process involving large fleets of connected vehicles. At the heart of this transition lies the problem of multi-vehicle dynamic navigation (MDN) [20, 39, 44], which aims to simultaneously route large fleets of vehicles under continuously evolving traffic conditions. Unlike single-vehicle navigation, MDN is a large-scale distributed coordination task, where each routing decision not only affects an individual vehicle but also alters the overall traffic flow across the network. This shift redefines the goal of navigation, moving beyond individual mobility optimization toward enhancing network-wide traffic efficiency. Despite recent advances, current approaches often fail to meet the demand of scalability, responsiveness, and cooperative decision-making, which are critical for real-world, city-scale deployment.

Existing approaches to vehicle navigation mainly fall into two paradigms. (1) *Shortest-path search methods* rely on classical algorithms such as Dijkstra [12] and A* [14], which compute optimal routes based on static or simplified cost functions. Their multi-vehicle extensions [1, 22, 23] attempt to coordinate fleets by injecting congestion penalties or capacity constraints. However, such methods remain largely rule-driven and depend on fixed assumptions about traffic dynamics. Thus, they struggle to handle

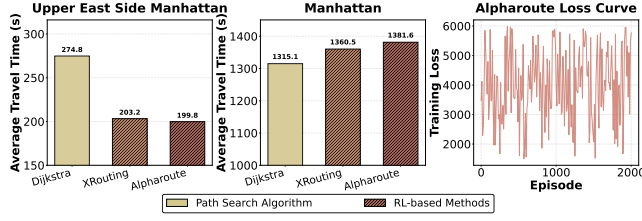


Figure 1: RL performance across different network scales. The Upper East Side covers about one-seventh of Manhattan.

the nonlinear, stochastic, and time-varying nature of real-world urban traffic [38], often producing brittle or inefficient routing outcomes. (2) *Reinforcement learning (RL) methods* model navigation as a sequential decision-making process, where models learn routing policies through interactions with simulated traffic environments [20, 33, 39, 44]. Although RL offers the advantage of adaptive learning and dynamic feedback, it suffers from fundamental scalability bottlenecks. As the number of vehicles or the size of the road network grows, the joint state-action space expands combinatorially, making efficient policy exploration and stable convergence extremely difficult. As shown in Figure 1, RL methods often perform competitively on small networks but experience sharp performance degradation at the city scale, sometimes even being outperformed by simple shortest-path baselines.

To overcome these limitations, we explore the potential of large language models (LLMs), which have recently demonstrated strong reasoning, flexible communication, and generalization capabilities across diverse domains [17, 24, 32]. Unlike traditional routing algorithms that rely on hand-crafted heuristics or RL methods that depend on costly trial-and-error training, LLMs provide a new foundation for adaptive, communication-driven coordination. With their vast pretrained knowledge and natural language reasoning capability, LLMs can integrate heterogeneous information, interpret evolving traffic conditions, and generate context-aware routing strategies. More importantly, LLMs inherently possess the ability to communicate and collaborate through language, allowing them to share information, negotiate priorities, and align local decisions with global objectives. Consequently, LLMs show great promise to overcome the scalability bottlenecks and coordination inefficiencies that constrain existing approaches.

Recent studies have begun to explore LLMs for multi-agent systems, showing that collaborative agents can decompose and solve complex tasks through structured communication. For example, MetaGPT [15] deploys four small-scale LLM agents organized into specialized roles to automate software development. In traffic management, CoLLMLight [45] employs up to five hundred agents leveraging spatiotemporal graphs for traffic signal coordination, while CoMAL [42] investigates small-scale collaborative traffic control scenarios with light traffic volumes. These works underscore the potential of LLM-driven coordination. However, unlike small-scale coordination, large-scale dynamic navigation involves continuous decision-making across tons of interacting vehicles, where both the *dimensionality of real-time data* and the *coupling between local and global dynamics* pose formidable challenges. Specifically, two critical questions emerge: (1) How can LLM agents exchange information and generate cooperative routing strategies while remain

computationally feasible at city scale? (2) How can we align each agent’s local routing decisions with the network-wide objective of minimizing congestion and maximizing overall mobility?

To tackle the above challenges, we propose CityNav, a scalable cooperative framework for network-wide dynamic navigation powered by LLM agents. CityNav adopts an efficient hierarchical architecture that integrates global coordination with local adaptability. At the macro level, a global traffic allocation agent aggregates network-wide information (e.g., congestion patterns and demand distributions), and formulates high-level directives to balance traffic loads across partitioned regions. At the micro level, a network of regional navigation agents manages routing within their assigned regions, generating intra-region paths that follow the global agent’s general directives while adapting to real-time local conditions. This division of responsibilities dramatically reduces complexity. The global agent focuses on strategic, network-wide decisions, while local agents handle smaller, localized navigation problems. By narrowing each agent’s decision space and enabling multi-level communication, CityNav achieves both scalability and responsiveness for large-scale multi-vehicle navigation.

To align local decisions with global optimization goals, CityNav introduces a cooperative reasoning optimization mechanism that jointly trains global and local LLM agents to balance per-vehicle efficiency and network-wide mobility. During training, agents interact with the traffic environment and receive real-time feedback on routing efficiency and congestion. Each agent is first optimized with an individual reward that reflects its task-specific objective (i.e., minimizing network congestion for the global traffic allocation and maximizing regional mobility for the local navigation). Then, both agents are jointly optimized with a shared network-wide reward, promoting coordination toward overall traffic efficiency. This hierarchical dual-reward optimization mechanism enables CityNav to achieve globally efficient yet locally adaptive navigation across large-scale urban road networks.

The key contributions of this work are threefold: (1) We propose CityNav, the first LLM-based multi-agent framework designed for city-scale dynamic navigation. By leveraging hierarchical coordination, CityNav enables scalable and adaptive routing across large road networks. (2) We design a cooperative reasoning optimization mechanism that explicitly aligns local decisions with global objectives. This approach fosters coordination among hierarchical agents, balancing individual performance with network-wide efficiency. (3) We conduct extensive experiments on four real-world road networks of varying scales (up to 1.6 million roads and 430,000 intersections) and traffic datasets, benchmarking CityNav against nine classical path search and RL-based methods. Results show consistent improvements in effectiveness, scalability, and computational efficiency.

2 Preliminary

Let the urban road network be represented as a directed graph $G = (V, E)$, where V is the set of intersections and E is the set of directed road segments. Traffic dynamics evolve over this network as vehicles move, interact, and affect congestion levels.

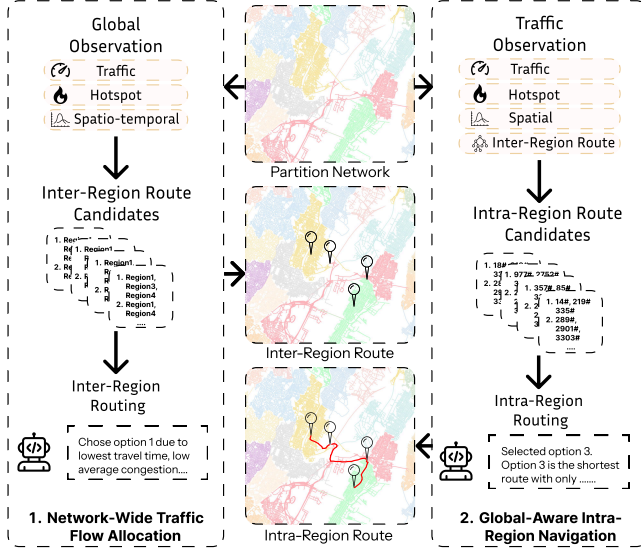


Figure 2: Workflow of CityNav framework.

Definition 2.1 (Multi-Vehicle Dynamic Navigation). The multi-vehicle dynamic navigation is formulated as a partially observable Markov decision process (POMDP), defined by a tuple $\langle S, O, A, P, R, N \rangle$:

- **Traffic state:** $s \in S$ denotes the traffic state, describing the joint condition of the road network;
- **Observation space:** $o \in O$ represents the partial observation available to navigation agents, where $o \subseteq s$;
- **Action space:** $a \in A$ corresponds to the routing decision, i.e., selecting the next region or road segment to traverse;
- **Routing plan:** P is the sequence of actions forming a vehicle's trajectory through regions or roads;
- **Reward function:** $r = f_{\text{reward}}(s, P)$ is the reward for navigation agents, where $P = \{P_1, \dots, P_N\}$ is the collection of routing plans.

Each vehicle travels from an origin node $v_1 \in V$ to a destination node $v_d \in V$ through a sequence of road segments while minimizing travel delay and avoiding congestion.

3 LLM for Multi-Vehicle Dynamic Navigation

As illustrated in Figure 2, the proposed *CityNav* framework introduces a hierarchical reasoning architecture that integrates global coordination with local adaptability to address large-scale multi-vehicle navigation. (1) At the upper level, a global traffic flow allocation agent oversees network-wide traffic conditions and dynamically formulates high-level routing strategies to balance flow distribution across different regions. (2) At the lower level, a set of local navigation agents operates within their assigned regions, where each agent generates fine-grained routing decisions tailored to local traffic dynamics. Guided by the global agent's strategic directives, local agents perform contextual reasoning to adapt to real-time variations, collaboratively ensuring efficient and congestion-aware vehicle navigation throughout the entire network.

3.1 Network-Wide Traffic Flow Allocation

At the upper level of *CityNav*, the global traffic allocation agent serves as the strategic coordinator that regulates network-wide traffic distribution. It aims to construct macro-level routing plans to avoid potential congestion. Before the operational loop begins, the entire road network is partitioned into a set of non-overlapping regions $\mathbf{z} = \{z_1, z_2, \dots, z_K\}$ using the Louvain community detection algorithm [5], which groups spatially and topologically cohesive areas while preserving strong intra-region connectivity.

3.1.1 Observation Construction. At each decision step, the traffic allocation agent receives an aggregated observation $o = \{o_{\text{traffic}}, o_{\text{hot}}, o_{\text{st}}\}$ composed of three categories of region-level features: (1) *Traffic conditions* (o_{traffic}): For each region $z_k \in \mathbf{z}$, we define a traffic descriptor:

$$o_{\text{traffic}}^{z_k} = [\text{cong}(z_k), \text{occ}(z_k), \bar{t}(z_k)], \quad (1)$$

where $\text{cong}(z_k)$, $\text{occ}(z_k)$, and $\bar{t}(z_k)$ denote the congestion level, occupancy rate, and average travel time within region z_k . (2) *Hotspot indicator* (o_{hot}): The current demand intensity of z_k is given by:

$$o_{\text{hot}}^{z_k} = \frac{N_{\text{active}}^k}{N_{\text{total}}}, \quad (2)$$

where N_{active}^k is the number of active vehicles whose routing plan include z_k . (3) *Spatio-temporal contexts* (o_{st}): Environmental context is captured by the set of congested edges E_{cong} and the current time index t , allowing the LLM to reason with temporal traffic priors and known spatial traffic bottlenecks.

3.1.2 Network-Wide Traffic Distribution. Given the observation o , the vehicle's origin region z_1 , and destination region z_d , we construct a traffic-aware prompt comprising: the current traffic observation o , the origin-destination pair $\langle z_1, z_d \rangle$, and a set of pre-screened, physically traversable candidate inter-region routes $\mathbf{P}_{\text{global}} = \{P_1, \dots, P_M\}$. Each route $P_m = \{z_1, \dots, z_d\} \subseteq \mathbf{z}$ represents an ordered sequence of regions connecting z_1 to z_d . Formally, the prompt is written as:

$$X_{\text{global}} = \text{Prompt}(o, z_1, z_d, \mathbf{P}_{\text{global}}). \quad (3)$$

The allocator's decision-making then proceeds in two stages, inspired by the ReAct framework [43]. First, during the reasoning phase, it analyzes the observation o , predicts emerging congestion patterns, and leverages its pre-trained knowledge of typical spatio-temporal dependencies to evaluate each candidate plan. Based on the reasoning, the allocation agent then selects the optimal route that minimizes expected travel delay and avoids congestion:

$$P_{\text{global}}^* = \pi_{\text{global}}^{\text{routing}}(\mathbf{P}, \pi_{\text{global}}^{\text{reason}}(X_{\text{global}})). \quad (4)$$

This two-stage process enables the global allocator to generate strategic, traffic-aware routing plans that are both interpretable and grounded in a comprehensive understanding of current and anticipated network-wide traffic conditions.

Operating at the lower level of *CityNav*, decentralized local navigation agents serve as tactical executors that translate the high-level strategic directives from the global traffic allocation agent into concrete, real-time routing plans within their assigned regions. The navigation is activated whenever a vehicle enters the assigned region of the agent, requesting an immediate control signal on which

road segment to traverse, while remaining aligned with the global inter-region routing plan P_{global}^* .

3.1.3 Observation Construction. To make context- and traffic-aware decisions, we construct a road segment-level observation $o' = \{o'_{\text{traffic}}, o'_{\text{hot}}, o'_{\text{spatial}}\}$ by aggregating three categories of local features: (1) *Traffic conditions* (o'_{traffic}): For each road segment e within the region, the traffic condition is described by:

$$o'_{\text{traffic}} = [\text{cong}(e), \text{occ}(e), \hat{\tau}(e)], \quad (5)$$

where $\text{cong}(e)$ is the congestion level, $\text{occ}(e)$ is the occupancy rate of the segment, and $\hat{\tau}(e)$ is the flow-free travel time (i.e., travel time without external delays such as congestion or red-light waiting). (2) *Hotspot indicator* (o'_{hot}): The immediate demand for boundary edges E_{bound} connecting the next target region z_{k+1} is quantified by $N_{\text{active}}^{\text{bound}}$, the number of vehicles currently routed toward road E_{bound} . (3) *Spatial contexts* (o'_{spatial}): This encodes the vehicle's current location e_i and the next region z_{k+1} in the global inter-region routing plan P_{global}^* , providing critical guidance to ensure local routing decisions are aligned with the network-wide strategy.

3.1.4 Global-Aware Route Planning. Firstly, we perform candidate path pre-screening, where all feasible paths from the vehicle's current location to the boundary roads are ranked by estimated free-flow travel time. The top ten paths form the candidate micro-routing plan set P_{local} , focusing the LLM agent's decision-making on the most promising options. Given the local observation o' , the target region z_{k+1} , and the candidate routing plan set P_{local} , the navigation agent executes a two-stage ReAct process, informed by the global intra-region plan P_{global}^* :

$$X_{\text{local}} = \text{Prompt}(o', z_{k+1}, P_{\text{local}}), \quad (6)$$

$$P_{\text{local}}^* = \pi_{\text{local}}^{\text{routing}}(P_{\text{local}}; \langle \pi_{\text{local}}^{\text{reason}}(X_{\text{local}}), P_{\text{global}}^* \rangle). \quad (7)$$

This hierarchical, context- and globally-aware routing mechanism ensures that intra-region decisions are both locally responsive and strategically aligned, enabling interpretable and efficient navigation across the entire traffic network.

3.1.5 Computational Complexity. Let $G = (V, E)$ denote the road network, z the set of partitioned regions where $|z| \ll |V|$, and V_r the intersections within region r where $|V_r| \approx |V|/|z|$. Traditional methods such as Dijkstra operate on the full graph with complexity $O(|E| + |V| \log |V|)$, while centralized frameworks like S-DTA require repeated simulation. CityNav decomposes routing into: (1) global planning across regions with $O(|z| \log |z|)$, and (2) local planning within each region with $O(|V_r| \log |V_r|)$. Since $|V_r| \ll |V|$, this reduces the dominant graph traversal cost by orders of magnitude.

4 Cooperative Reasoning Optimization

While the hierarchical CityNav framework effectively reduces the complexity of network-wide vehicle navigation, off-the-shelf LLMs are not explicitly designed for coordinated, top-down vehicle navigation control. To address this, we introduce a cooperative reasoning optimization mechanism that enables agents to learn from the collective impact of their mutual routing decisions. Within this paradigm, the global traffic allocator and decentralized local navigation agents are jointly trained to reason toward a shared system-level

objective, ensuring that local routing decisions remain globally consistent. This design allows CityNav to maintain network-wide efficiency while preserving the flexibility under real-time dynamics.

4.1 Coordination Reward Construction

To facilitate cooperation across hierarchical agents, we design a dual-reward mechanism composed of an individual reward that promotes per-vehicle efficiency and a shared reward that aligns all agents toward global traffic optimization.

The individual reward quantifies per-vehicle efficiency while maintaining fairness across routes with different traffic contexts:

$$r_{\text{ind}} = \frac{\tau}{\tau + \hat{\tau}} - \lambda \cdot N_{\text{idle}}, \quad (8)$$

where τ is the actual travel time, $\hat{\tau}$ is the flow-free travel time, N_{idle} is the number of idle steps, and λ controls the penalty for prolonged idling. This design ensures fairness: the ratio $\tau/(\tau + \hat{\tau})$ evaluates vehicles relative to their own baseline, while α balances individual fairness against network-wide efficiency.

To enable cooperative reasoning between two hierarchical agents, we introduce a shared reward that reflects network-wide mobility:

$$r_{\text{share}} = -\frac{1}{|P_{\text{global}}^*|} \sum_{z \in P_{\text{global}}^*} \bar{\tau}(z), \quad (9)$$

where $P_{\text{global}}^* = \{z_1, \dots, z_d\}$ denotes the sequence of regions in the global routing plan, and $\bar{\tau}(z)$ is the average travel time of vehicles traversing region z . This formulation provides a unified optimization signal: the global traffic allocation agent learns to minimize aggregated regional delays, while local navigation agents adapt their routing to support the global plan.

4.2 Bi-Level Reasoning Optimization

Building upon this reward structure, we develop a bi-level reasoning optimization mechanism that jointly post-trains the global traffic allocation and local navigation agents toward a cooperative equilibrium. This framework ensures that reasoning processes across both levels are optimized to balance per-vehicle efficiency and network-wide coordination.

At each training iteration, for each agent, we sample N reasoning processes under the traffic state s :

$$\mathcal{D} = \{y_1, y_2, \dots, y_N\} \sim \pi_{\theta}(\cdot | s), \quad (10)$$

where each reasoning process y_i represents a full reasoning-action sequence generated by policy π_{θ} . Each reasoning process receives a scalar reward combining individual and overall objectives:

$$r_i = \alpha r_{\text{ind}}^i + (1 - \alpha) r_{\text{share}}^i, \quad (11)$$

where $\alpha \in [0, 1]$ controls the trade-off between per-vehicle efficiency and network-wide coordination.

Inspired by [35], we adopt a multi-agent group reinforcement policy optimization (GRPO) [13] process to jointly optimize the two hierarchical agents. To stabilize optimization, we compute a group-normalized advantage without the critic network:

$$A_i = r_i - \bar{r}, \quad \text{where} \quad \bar{r} = \frac{1}{N} \sum_{j=1}^N r_j. \quad (12)$$

Table 1: Statistics of datasets.

Dataset	Intersections/Roads	Vehicles	Arrival rate (vehicles/5min)			
			Mean	Std	Max	Min
NYC	429,581/1,639,481	1,038,608	3,570.28	1,359.76	10,675	999
Manhattan	18,240/74,979	24,880	86.39	80.04	332	12
UES	3,955/11,134	2,000	27.03	2.16	28	9
Chicago	249,511/978,799	79,342	596.56	9.42	600	490

The positive advantage indicates the reasoning process outperforms the group average, directly guiding policy improvement. The moving average of $\bar{r}$ is optionally used to smooth variance across training steps. For each reasoning process, the clipped surrogate loss is then defined as:

$$L_{\text{clip}}^i(\theta) = \min \left(\frac{\pi_{\theta}(y_i | s)}{\pi_{\theta_{\text{ref}}}(y_i | s)} A_i, \text{clip} \left(\frac{\pi_{\theta}(y_i | s)}{\pi_{\theta_{\text{ref}}}(y_i | s)}, 1 - \epsilon, 1 + \epsilon \right), A_i \right), \quad (13)$$

where ϵ is the clipping parameter that stabilizes policy updates. The overall cooperative GRPO objective jointly optimizes both agents' parameters $\Theta = \{\theta_{\text{global}}, \theta_{\text{local}}\}$ while enforcing alignment through a KL penalty:

$$\max_{\Theta} \mathbb{E}_{i \sim \mathcal{D}} \left[\sum_{\ell \in \{\text{global}, \text{local}\}} \left(\frac{1}{N} \sum_{i=1}^N L_{\text{clip}}^i(\theta_{\ell}) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta_{\ell}} \parallel \pi_{\theta_{\text{ref}}^{\ell}}) \right) \right], \quad (14)$$

where $\Theta = \{\theta_{\text{global}}, \theta_{\text{local}}\}$ are model parameters of global and local agents, and β is the KL coefficient controlling update stability.

During training, both agents are optimized asynchronously using the AdamW optimizer with progressive learning rate schedules. Loss masking is applied to exclude observation tokens from gradient computation, along with mixed-precision (AMP) training and gradient clipping. Through this cooperative bi-level optimization, CityNav progressively aligns the reasoning and policies of hierarchical LLM agents, achieving locally adaptive yet globally consistent network-wide vehicle navigation.

5 Experiments

In this section, we conduct extensive experiments to evaluate our proposed CityNav by answering the following research questions:

- How is the **effectiveness** of CityNav compared with traditional path search algorithms and RL-based methods?
- How is the **scalability** ability of CityNav across different scales of road networks and traffic volumes?
- How is the **generalization** of CityNav in unseen road networks and traffic contexts?

5.1 Experimental Settings

5.1.1 Datasets. We conduct our experimental evaluation on four real-world urban traffic datasets that differ in both spatial scale and network topology. The origin–destination (OD) data for controlled vehicles are derived from the New York City and Chicago Taxi datasets [9, 28], which capture fine-grained mobility demand patterns across time and space. A randomly selected 2% of trips is designated as optimization targets managed by our framework. To faithfully reconstruct realistic city-scale traffic conditions, the

remaining 98% of trips, along with additional background vehicle volumes, are calibrated using official mobility surveys [27]. Following [47], A gravity-based model [2] is deployed for inter-region OD demand generation, where the background traffic flow between two regions is proportional to their activity intensities.

- **New York City (NYC):** The complete road network of New York City serves as the primary experimental environment. It comprises 429,581 intersections and captures diverse traffic dynamics across urban, suburban, and peripheral areas.
- **Manhattan:** The Manhattan sub-network, consisting of 18,240 intersections, represents a dense urban core characterized by recurrent congestion, grid-like topology, and complex interactions between arterial and local roads.
- **Upper East Side (UES):** The Upper East Side, a compact sub-region within Manhattan, NYC, containing 3,955 intersections, provides a small-scale setting for evaluating localized navigation and coordination behaviors.
- **Chicago:** Chicago, with 249,511 intersections, is used to evaluate the zero-shot generalization ability of our framework across distinct city layouts and mobility distributions.

5.1.2 Data Anonymization. The taxi trip data used in our experiments is sourced from publicly available NYC and Chicago Open Data [8, 10]. The dataset is anonymized and does not contain any personally identifiable information regarding drivers or passengers.

5.1.3 Environment Settings. All experiments are conducted using SUMO [25], a widely adopted open-source microscopic traffic simulator that enables high-fidelity modeling of vehicular dynamics and network-level interactions. SUMO supports large-scale simulations with realistic mobility and provides an interactive visualization interface for monitoring traffic states. It offers diverse APIs for retrieving traffic features and implementing external control policies.

5.2 Metrics

Following prior studies [21, 26, 44], we evaluate the performance of baseline methods using four key traffic efficiency metrics: average travel time (ATT), average waiting time (AWT), average delay time (ADT), and throughput (TP). These metrics jointly capture both individual-level mobility efficiency and overall network throughput.

- **Throughput (TP):** The total number of vehicles that successfully reach their destinations within the simulation period. Higher values indicate better overall throughput.
- **Average Travel Time (ATT):** The mean duration of all completed trips, measured from origin to destination. Lower values indicate more efficient routing.
- **Average Waiting Time (AWT):** The mean total time vehicles remain stationary in the network (e.g., at intersections or due to congestion). Lower values are better.
- **Average Delay Time (ADT):** The average deviation between a vehicle's actual travel time and its theoretical free-flow travel time. Lower values reflect smoother network operations.

5.3 Compared Approaches

To comprehensively evaluate the effectiveness of CityNav, we benchmark it against nine representative baselines. For path search algorithms, we select: **Dijkstra** [12] serves as the classical deterministic baseline, computing globally optimal shortest paths based on static network topology and road length; **MinDits** adopts a greedy strategy that selects the next-hop road closest to the destination in terms of network distance. **MinLat** follows a similar greedy heuristic but chooses the next-hop road with the smallest estimated travel latency based on predicted traffic conditions using ARIMA [6]; **SBP** [23], which optimizes continuous route combinations; **DyAM** [22], which enforces routing through dynamic avoidance maps; and **S-DTA** [1], a meta-heuristic framework for simulation-based dynamic traffic assignment. For RL-based methods, we include: **XRouting** [39], an early deep reinforcement routing method; **Adaptation-Navigation** [3], which emphasizes adaptive policy learning under non-stationary traffic conditions; and **AlphaRoute** [26], the SOTA multi-agent coordination framework.

5.4 Model Settings

We adopt Qwen-3-8B [41] as the backbone LLM, which offers strong performance across diverse tasks while maintaining a lightweight model size. To ensure stable and reproducible training, a consistent set of hyperparameters is used across all experiments. A conservative learning rate of 1×10^{-4} is applied, with a replay buffer capacity of 10,000 experiences. Training follows a mini-batch strategy with a base batch size of 4 and 8 steps of gradient accumulation. For the LLM configuration, we set `top_p` as 1.0 and the temperature to 0.1. Low-Rank Adaptation (LoRA) fine-tuning is employed with a rank of 8 and a scaling factor $\alpha = 16$. Post-training data is collected from interactive simulations on the NYC dataset.

5.5 Comparison Between the Traditional Methods and CityNav (RQ1 & RQ2)

We first conduct experiments on three different scales of real-world traffic scenarios, including NYC, Manhattan, and Upper East Side, to evaluate the effectiveness and scalability of our proposed framework. The comparison between traditional and learning-based approaches is summarized in Table 2.

CityNav demonstrates significant advantages in diverse scales of road networks, achieving the highest throughput with consistently low ATT, AWT, and ADT. On the New York City dataset, the largest and most complex scenario, CityNav successfully completes nearly three thousand trips, achieving more than four times the throughput of the best competing approach while maintaining efficient travel and delay times. In contrast, the path-search algorithms (e.g., Dijkstra, MinDits, and S-DTA) suffer from severe congestion, completing only about 21% of the throughput achieved by CityNav. Although some of these methods appear to report lower ATT, most vehicles remain stuck in traffic congestion and fail to reach their destinations, resulting in relatively low ATT that reflects only a small subset of successful trips. Notably, no RL-based method converged in this city-scale scenario due to the combinatorial explosion of the state-action space, while CityNav's hierarchical decomposition enables stable convergence.

On the smaller Upper East Side network, recent path search algorithms (e.g., SBP and S-DTA) and RL-based methods (e.g., Alpharoute) achieve strong performance, benefiting from manageable state and action spaces. However, as the network scale increases to the full Manhattan and New York City settings, their performance deteriorates sharply or becomes infeasible to train due to the exponential growth of environment complexity. Conversely, CityNav maintains robust scalability, effectively managing network-wide coordination and congestion even under diverse traffic settings.

Overall, these results confirm that CityNav delivers globally efficient and congestion-resilient traffic management across varying network scales. Our proposed method outperforms both path-search and RL-based navigation baselines while maintaining strong scalability to complex, real-world traffic environments.

5.6 Traffic Demand Scalability Comparison (RQ2)

A key requirement for any city-scale navigation framework is the ability to scale effectively under increasing traffic demand. To examine this property, we evaluate all methods on the New York City dataset by varying the proportion of optimized trips from 2% to 10% of total origin–destination (OD) demand during the morning peak hours (8 AM–12 PM). This setting simulates a progressively denser traffic environment, where the number of vehicles subject to active coordination rises fivefold. As shown in Table 3, higher demand naturally leads to increased congestion, reflected by longer average travel time (ATT), average waiting time (AWT), and average delay time (ADT) across all systems.

From the results, CityNav demonstrates remarkable robustness and scalability. When the proportion of optimized vehicles increases fivefold, throughput (TP) grows steadily from 1,022 to 4,294 while maintaining moderate increases in ATT, AWT, and ADT relative to the traffic surge. This stability suggests that CityNav's cooperative, bi-level reasoning effectively coordinates vehicles to mitigate localized congestion and sustain network-wide flow efficiency. The results confirm that CityNav generalizes well across different traffic intensities, preserving both global coordination stability and local adaptability even in heavy-demand scenarios. In terms of computational overhead, we observe that runtime scales primarily with fleet size rather than map size. For real-world deployment in a full metropolitan environment such as New York, we estimate that $16 \times$ NVIDIA H800 GPUs would reduce end-to-end inference time to approximately 3 seconds.

5.7 Generalization Comparison (RQ3)

To evaluate the model's ability to generalize beyond its training environment, we conducted a zero-shot transfer experiment. Models trained on the New York City road network were directly deployed on the Chicago network, which differs substantially in both topology and traffic dynamics. This setup tests whether each method learns transferable routing principles or merely overfits to the specific spatial and flow characteristics of the training environment. The results are summarized in Table 4.

CityNav demonstrates strong generalization capability, achieving a throughput of 1,073 successfully routed vehicles, approximately twice that of the best-performing path search algorithm,

Table 2: Overall performances of CityNav and traditional methods on New York City, Manhattan, and Upper East Side datasets. The symbol “-” indicates that the corresponding model failed to converge during training on the given dataset.

Methods	New York City				Manhattan				Upper East Side			
	TP ↑	ATT ↓	AWT ↓	ADT ↓	TP ↑	ATT ↓	AWT ↓	ADT ↓	TP ↑	ATT ↓	AWT ↓	ADT ↓
Path Search Algorithms												
Dijkstra	645	703.83	98.88	36785.70	378	1315.14	23.93	1002.03	40	274.82	42.10	167.04
MinDits	632	810.48	98.89	36889.75	356	1370.99	24.88	1021.35	40	276.68	43.23	186.92
MinLat	632	810.47	98.89	36889.79	356	1370.79	24.58	1096.39	40	276.68	43.23	176.20
SBP	488	10874.38	91.23	20480.81	193	1140.50	24.27	789.99	40	255.35	39.80	137.70
DyAM	778	10759.85	92.57	20332.25	262	1072.23	23.29	693.49	40	275.87	37.08	145.74
S-DTA	758	645.43	99.19	35669.84	257	1033.98	49.44	730.36	40	255.08	107.23	156.49
RL-based Methods												
Xrouting	-	-	-	-	217	1360.48	33.37	1091.13	40	203.24	43.23	107.65
Adaptation-Navigation	-	-	-	-	241	1599.33	36.05	1255.24	40	230.89	21.47	134.39
Alpharoute	-	-	-	-	225	1381.64	48.85	1083.87	40	199.78	7.78	104.92
CityNav	2952	2771.21	89.34	4207.46	393	667.99	30.09	357.79	40	257.52	38.93	135.29

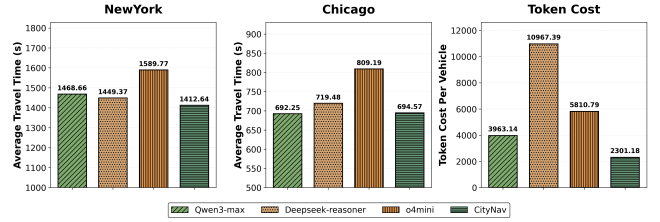
Table 3: Performance in different scales of traffic demand.

Datasets	Ratio	Metrics			
		TP	ATT	AWT	ADT
New York	2%	1022	1412.64	35.38	2019.56
	5%	2489	1975.57	47.77	2722.46
	10%	4294	2922.81	68.43	3871.39

Table 4: Zero-shot experiment on the Chicago dataset.

Datasets	Methods	Metrics			
		TP	ATT	AWT	ADT
Chicago	Dijkstra	595	419.57	78.48	14388.15
	MinDits	683	460.92	98.95	14265.66
	MinLat	683	482.25	98.96	14288.87
	SBP	758	4897.45	60.33	12859.95
	DyAM	788	4996.85	76.52	12796.56
	S-DTA	655	475.31	68.42	16745.71
	CityNav	1073	1609.40	58.56	2561.78

while maintaining considerably lower ATT, AWT, and ADT. This shows that CityNav effectively transfers its learned coordination strategies to an unseen city without retraining, preserving both routing efficiency and network stability under new conditions. In contrast, the path-search algorithms (e.g., Dijkstra, S-DTA, and MinDits) suffer from severe congestion, resulting in a drastic reduction in throughput as most vehicles become trapped in traffic and only a few short trips complete successfully. Overall, these results confirm that CityNav not only scales efficiently with increasing traffic volumes but also generalizes robustly across heterogeneous urban topologies, achieving consistent, congestion-resilient performance even in unseen road network environments.

**Figure 3: Comparison between leading LLMs and CityNav.**

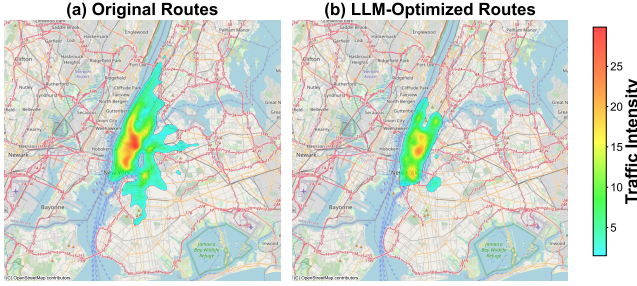
5.8 Comparison with Leading LLMs

This final experiment examines the significance of CityNav’s domain-specific fine-tuning by replacing its backbone model with several state-of-the-art general-purpose LLMs, including Qwen3-Max (over 1T parameters), DeepSeek-R1 (761B), and o4-mini (close-source). Evaluations were conducted on both the NYC dataset for in-domain evaluation and the Chicago dataset for zero-shot generalization examination during the morning rush hour (8 AM–12 PM). The results are presented in Figure 3.

Despite being built upon a lightweight 8B-parameter backbone, CityNav consistently achieves the lowest ATT across both datasets, matching or surpassing much larger off-the-shelf LLMs. This superior performance arises from CityNav’s bi-level cooperative fine-tuning, which enable coordinated and adaptive reasoning under dynamic congestion conditions. Moreover, CityNav attains these results with the least token usage (58.06% of the tokens used by Qwen3-Max), owing to its hierarchical reasoning structure that minimizes redundant inference and communication among agents. This exceptional efficiency–performance balance underscores CityNav’s scalability and practicality. This demonstrates that our proposed targeted domain optimization effectively surpasses generic large-scale reasoning models in both effectiveness and computational efficiency in real-world deployment scenarios.

Table 5: Ablation tests.

Datasets	Methods	Metrics			
		TP	ATT	AWT	ADT
NewYork	w/o RL	1883	6762.54	53.25	9324.45
	w/o hierarchy	493	9983.78	93.37	18853.75
	CityNav	2952	2771.21	89.34	4207.46

**Figure 4: Traffic coordinated by Dijkstra and CityNav.**

5.9 Ablation Study

To evaluate the contributions of CityNav’s key components, we conducted ablation experiments on two primary variants: (1) CityNav w/o hierarchy, where the multi-level agent structure is collapsed into a single centralized agent responsible for all routing decisions, testing the necessity of hierarchical decomposition for handling city-scale state spaces; and (2) CityNav w/o RL, which retains the hierarchy but omits the cooperative reasoning optimization, isolating the effect of reinforcement learning on coordination. All three versions, including the full CityNav, were evaluated on the NYC dataset from 0 AM to 12 PM. The results are summarized in Table 5.

The results demonstrate that both the hierarchical framework and RL-based reasoning optimization are essential for CityNav’s performance. Without the hierarchy, the centralized agent fails to efficiently capture network-wide traffic conditions while managing individual vehicle routing, leading to severe bottlenecks and drastically reduced throughput. Removing RL weakens the coordination between global traffic allocation and local navigation agents, resulting in suboptimal routing and higher travel times. In contrast, the full CityNav model effectively balances per-vehicle efficiency with global traffic mobility, achieving substantially lower ATT, AWT, and ADT while completing nearly twice as many trips. These findings confirm that CityNav can robustly handle city-scale traffic optimization with efficient multi-agent coordination.

5.10 Case Study

To provide a macroscopic and intuitive view of CityNav’s impact on urban traffic, we conducted a case study visualizing traffic distribution across the New York City road network. In Figure 4, we show traffic heatmaps comparing routing coordinated by Dijkstra versus optimized by CityNav, with color intensity indicating the number of vehicle passes on each road segment.

The comparison demonstrates that CityNav significantly mitigates congestion across the city. Before deployment, heavy traffic accumulations were concentrated throughout central areas. After

deploying CityNav, congestion in the hotspots (*i.e.*, regions in red) is markedly reduced, while traffic flow in the rest of the network becomes smoother and more evenly distributed. Overall, CityNav promotes a more balanced and efficient utilization of urban road capacity, effectively alleviating network-wide congestion while maintaining high throughput and travel efficiency.

6 Related Work

6.1 Vehicle Navigation

Early routing utilized static algorithms and dynamic assignment [30], while recent deep RL approaches target congestion-aware navigation [31]. Asynchronous and large-scale MARL frameworks effectively handle dynamic routing and congestion smoothing [11, 44]. Additionally, hybrid RL–DTA methods combine system-optimal principles with learning to enhance equilibrium stability [37, 46], supporting resilient city-level management.

6.2 Intelligent Transportation

Scalable urban control increasingly adopts hierarchical frameworks to align global coordination with local adaptability. Systems like NavTL [?] and HiLight [40] use multi-level structures to coordinate regional strategies with intersection control. Such hierarchical MARL systems outperform flat schemes in stability and efficiency by balancing top-down optimization with bottom-up adaptability.

6.3 Large Language Models for Decision Making

LLMs serve as cognitive agents for reasoning and coordination [4, 7, 19, 34]. Frameworks like ReAct [43], MetaGPT [15], and generative agents [29] demonstrate capabilities in interleaved reasoning and emergent coordination, while AgentsCoDriver [16] applies LLMs to vehicle communication. These advances enable deliberative spatiotemporal reasoning, directly informing CityNav’s hierarchical design.

7 Conclusion and Future Work

This study proposes CityNav, the first LLM-based hierarchical framework for large-scale dynamic navigation. By coupling a global traffic allocation agent with decentralized local agents, CityNav effectively bridges global coordination and local adaptability. Its cooperative reasoning mechanism aligns local decisions with global objectives to balance individual efficiency and network mobility. Experiments on real-world networks demonstrate that CityNav outperforms classical and reinforcement learning baselines in efficiency, congestion reduction, and scalability. Future work will integrate multimodal factors, such as pedestrians and public transportation, to further advance intelligent urban mobility coordination.

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A Appendix

A.1 Prompt Template

We present the prompt template of global traffic allocation and local navigation agents in figure 5 and figure 6.

```
Macro Route Planning: Region 12 -> Region 4
Vehicle:
vehicle_id: trip.57662
start_region: 12, dest_region: 4
Region_State_Snapshot:
R0: vehicles=9, avg_speed=9.8m/s, cong=1.20, capacity_util=0.62
R1: vehicles=10, avg_speed=9.1m/s, cong=1.80, capacity_util=0.70
R2: vehicles=18, avg_speed=5.5m/s, cong=2.80, capacity_util=0.78
R3: vehicles=8, avg_speed=10.2m/s, cong=1.50, capacity_util=0.65
R4: vehicles=23, avg_speed=4.8m/s, cong=3.20, capacity_util=0.85
R6: vehicles=12, avg_speed=8.3m/s, cong=2.10, capacity_util=0.68
R12: vehicles=15, avg_speed=7.5m/s, cong=2.30, capacity_util=0.72
R18: vehicles=7, avg_speed=11.0m/s, cong=1.20, capacity_util=0.58
R19: vehicles=28, avg_speed=3.2m/s, cong=3.80, capacity_util=0.92
R27: vehicles=31, avg_speed=2.8m/s, cong=4.10, capacity_util=0.95
Routes:
Rt1: R12>R3>R4 | cong_series: R12:2.30|R3:1.50|R4:3.20 | cap_series: R12:0.72|
R3:0.65|R4:0.85 | speed_series: R12:7.5|R3:10.2|R4:4.8
Rt2: R12>R6>R2>R18>R4 | cong_series: R12:2.30|R6:2.10|R2:2.80|R18:1.20|R4:3.20 |
cap_series: R12:0.72|R6:0.68|R2:0.78|R18:0.58|R4:0.85 | speed_series: R12:7.5|R6:8.3|
R2:5.5|R18:11.0|R4:4.8
Rt3: R12>R6>R2>R1>R4 | cong_series: R12:2.30|R6:2.10|R2:2.80|R1:1.80|R4:3.20 |
cap_series: R12:0.72|R6:0.68|R2:0.78|R1:0.70|R4:0.85 | speed_series: R12:7.5|R6:8.3|
R2:5.5|R1:9.1|R4:4.8
Rt4: R12>R6>R2>R3>R4 | cong_series: R12:2.30|R6:2.10|R2:2.80|R3:1.50|R4:3.20 |
cap_series: R12:0.72|R6:0.68|R2:0.78|R3:0.65|R4:0.85 | speed_series: R12:7.5|R6:8.3|
R2:5.5|R3:10.2|R4:4.8
Coordination:
overloaded: R19(cong:3.80,cap:0.92), R27(cong:4.10,cap:0.95),
R4(cong:3.20,cap:0.85), R2(cong:2.80,cap:0.78)
available: R0(cong:1.20,cap:0.62), R18(cong:1.20,cap:0.58), R3(cong:1.50,cap:0.65)
Cut-Edge_Status:
BE_15_2: vehicles=5, cong=2.30, util=0.68, flow_rate=180.0veh/h
BE_15_2: vehicles=5, cong=2.30, util=0.68, flow_rate=180.0veh/h
BE_2_15: vehicles=3, cong=1.80, util=0.55, flow_rate=120.0veh/h
BE_6_2: vehicles=4, cong=2.10, util=0.62, flow_rate=150.0veh/h
BE_6_2: vehicles=4, cong=2.10, util=0.62, flow_rate=150.0veh/h
BE_14_0: vehicles=2, cong=1.50, util=0.48, flow_rate=90.0veh/h
BE_3_4: vehicles=6, cong=2.50, util=0.75, flow_rate=210.0veh/h
BE_12_3: vehicles=4, cong=1.80, util=0.60, flow_rate=135.0veh/h
BE_18_4: vehicles=3, cong=1.50, util=0.52, flow_rate=105.0veh/h
Objective: Select route based on current raw regional and boundary metrics; avoid
known congested areas and stuck edges.
ROUTE_OPTIONS: [12, 3, 4]/[12, 6, 2, 18, 4]/[12, 6, 2, 1, 4]/[12, 6, 2, 3, 4]
GLOBAL_GUIDANCE:
Goals: minimize system-wide travel time, balance regional load, avoid severe
congestion
AvoidRegions: R19,R27
AvoidEdges: 1328669510#6,633946076,1049318472#0,235916944#0
Note: Morning rush hour detected; R4 and R2 showing elevated congestion.
Consider alternative routes through R3 or R18.
Key_Regional_Congestion: R0:1.2|R1:1.8|R2:2.8|R3:1.5|R4:3.2|R6:2.1|R12:2.3|R18:1.2|
R19:3.8|R27:4.1
```

Figure 5: Global traffic allocation prompt.

REGIONAL ROUTE PLANNING FOR VEHICLE trip.57662
 Current region: 12, Target region: 4
 Current edge: 173393205#14
 Current time: 28800.0s
 OBJECTIVE: Avoid congestion and hotspots; do not pick only lowest travel time.
 GUIDELINES: Prefer higher avg speed edges, lower congestion; avoid edges with known exit hamper or hotspot events.
 CONSIDER: Regional traffic status and other vehicles heading to same destination; choose a robust route under current conditions.
 ROUTE CANDIDATES:
 Option 1: Route to -1065132315#6 | Length: 68 edges | AvgCongScore: 0.07 | BoundaryCongScore: 0.70
 Time: 217.3s | Dist: 5224.6m | EdgeCong[0..]: 173393205#14:1.2|34075306#38:0.8|34075306#39:1.5|-1101320094#0:2.1|-435545055#1:2.8|-435545059#0:1.9|-1101320095:1.3|-435397268#4:0.9 | EdgeSpeed[0..]: 173393205#14:8.5|34075306#38:10.2|34075306#39:7.8|-1101320094#0:5.2|-435545055#1:3.8|-435545059#0:6.1|-1101320095:9.2|-435397268#4:10.8 | AvgCong: 1.52 | AvgSpeed: 7.8m/s | VehTotal: 23 | BoundaryCong: 2.30 | PlannedUse: 7 | Edges: 68
 Option 2: Route to -231603332#3 | Length: 10 edges | AvgCongScore: 0.21 | BoundaryCongScore: 0.70
 Time: 55.9s | Dist: 844.8m | EdgeCong[0..]: 173393205#14:1.2|34075306#38:0.8|34075306#39:1.5|1101320092#1:2.3|435545058#0:2.1|435545058#1:1.7|231603329#0:1.4|231603329#1:1.9 | EdgeSpeed[0..]: 173393205#14:8.5|34075306#38:10.2|34075306#39:7.8|1101320092#1:5.8|435545058#0:6.2|435545058#1:7.5|231603329#0:8.8|231603329#1:7.1 | AvgCong: 1.61 | AvgSpeed: 7.7m/s | VehTotal: 12 | BoundaryCong: 1.80 | PlannedUse: 4 | Edges: 10
 Option 3: Route to -231603339#7 | Length: 19 edges | AvgCongScore: 0.11 | BoundaryCongScore: 0.70
 Time: 108.7s | Dist: 1422.8m | EdgeCong[0..]: 173393205#14:1.2|34075306#38:0.8|24112868#0:1.6|24112868#4:1.9|-401896407#3:2.5|-401896407#2:1.8|787322380:1.2|-435720295#1:0.9 | EdgeSpeed[0..]: 173393205#14:8.5|34075306#38:10.2|24112868#0:7.9|24112868#4:6.8|-401896407#3:4.9|-401896407#2:7.2|787322380:9.5|-435720295#1:10.8 | AvgCong: 1.48 | AvgSpeed: 8.2m/s | VehTotal: 15 | BoundaryCong: 1.60 | PlannedUse: 3 | Edges: 19
 REGION TRAFFIC: avgCong=2.30, vehicles=15
 BOUNDARIES: -1065132315#6 cong=2.3 used=7 | -231603332#3 cong=1.8 used=4 | -231603339#7 cong=1.6 used=3
 GLOBAL GUIDANCE:
 Goals: minimize system-wide travel time, balance regional load, avoid severe congestion
 AvoidEdges:
 1328669510#6,633946076,1049318472#0,235916944#0,-1413118082#3,-435536164#1,435545058#0,1023289873#0
 Note: Morning rush hour detected; prefer routes with higher average speeds and lower peer pressure on boundaries.
 PEERS_TO_R4: 8 vehicles also heading to Region 4
 PEER_CHOICES: -1065132315#6:7 | -231603332#3:4 | -231603339#7:3
 REGION 12: 15 vehicles, 72 boundaries
 High util: -1065132315#6:7, 34075306#38:5, 231603329#0:4
 DECISION FORMAT: Reply with the option number only (1.5).
 REMINDER: Do NOT rely on travel time alone. Use congestion/speed and peer pressure to avoid jams.

Figure 6: Local navigation prompt.